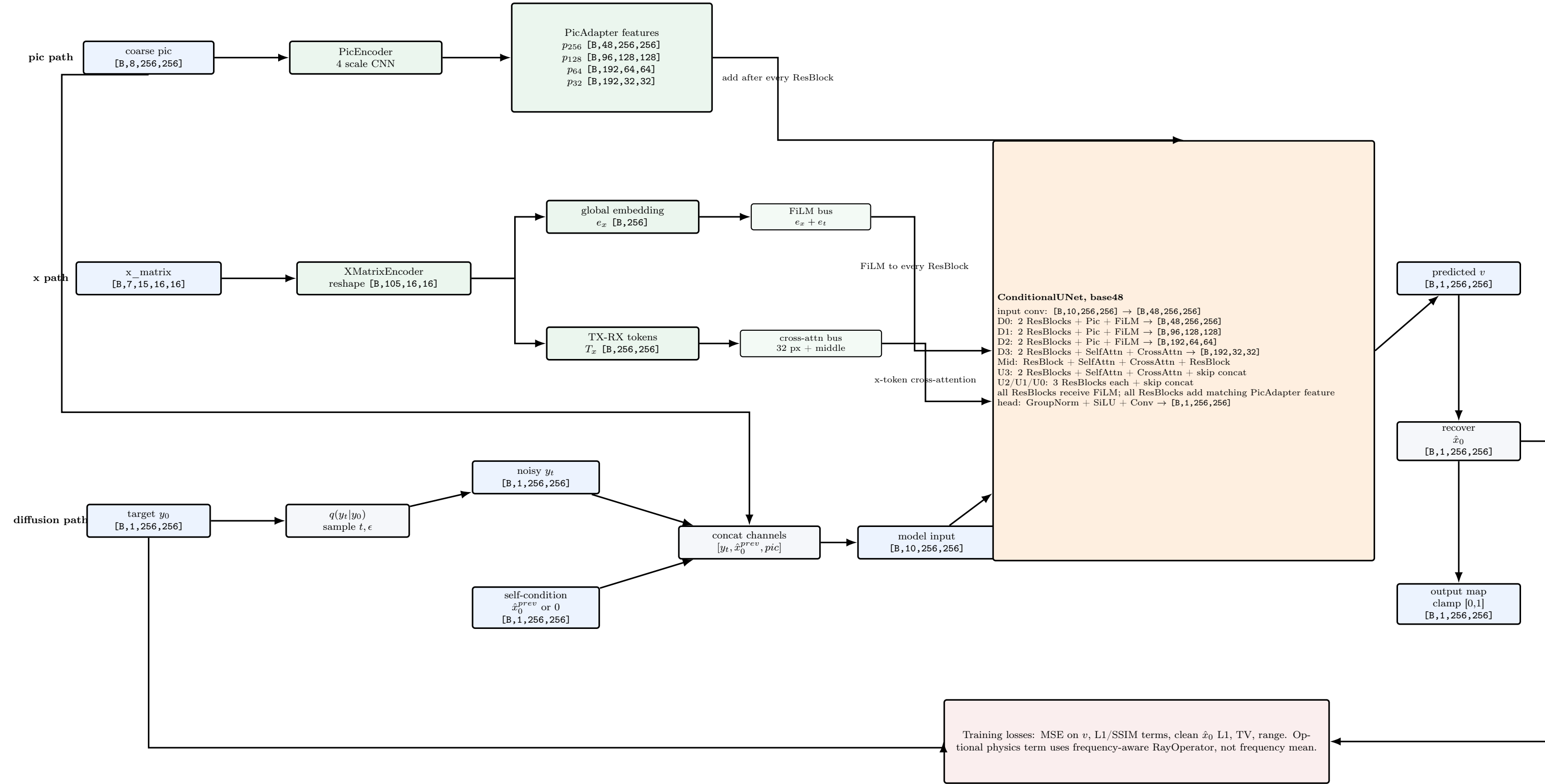


# Current Conditional Diffusion Network for Ultrasonic Defect Map Inversion

Follows `dataset_a_256_base48.yaml`: image size 256, pic channels 8, x channels 7, frequencies 15, base channels 48.  
Tensor order: image tensors use `[B,C,H,W]`; x-matrix uses `[B,C,F,TX,RX]`; x tokens use `[B,N,D]`.

**Training view.** The diffusion variable is the continuous target label  $y_0$ . The network sees the noisy label  $y_t$ , the previous clean prediction used for self-conditioning, the 8-channel coarse-map condition, and the full frequency x-matrix.

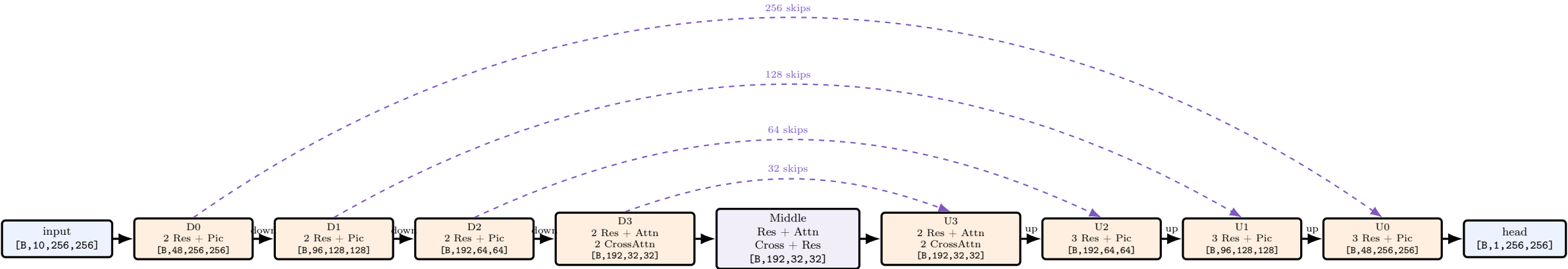
**Sampling view.** Start from Gaussian noise `[B,1,256,256]`. At each DDPM/DDIM step, feed the previous  $\hat{x}_0$  as self-condition. Optional physics guidance adjusts  $\hat{x}_0$  through the frequency-aware RayOperator.



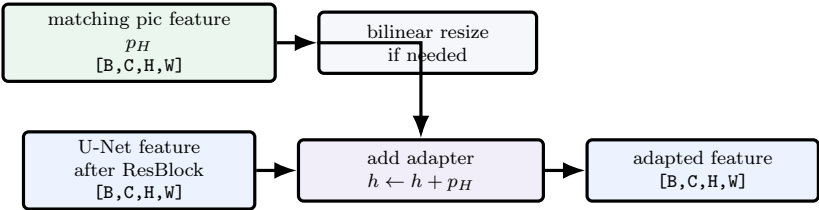
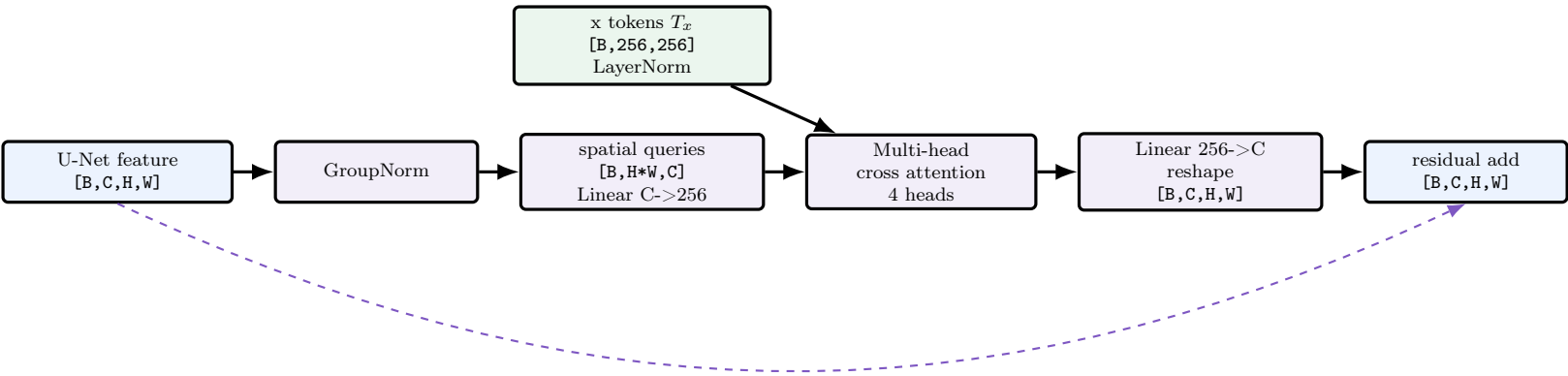
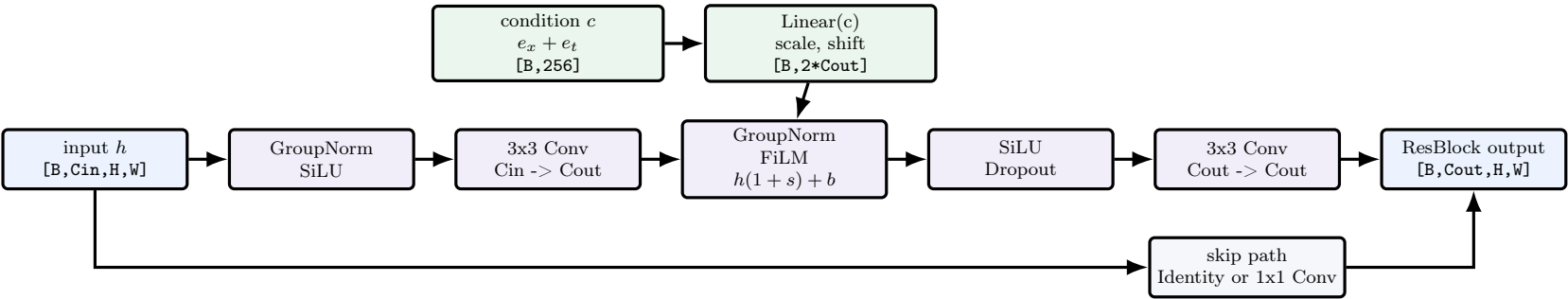
U-Net Tensor Shapes and Block Counts

Configuration: base=48, channel\_mult=[1,2,4,4], num\_res\_blocks=2, attention\_resolutions=[32], cross\_attention\_resolutions=[32].

| Stage        | Operation                                           | Input shape                                         | Output shape    | Notes                                                                                           |
|--------------|-----------------------------------------------------|-----------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------|
| Input concat | concat $y_t$ , self-condition, pic                  | [B,1,256,256] +<br>[B,1,256,256] +<br>[B,8,256,256] | [B,10,256,256]  | Self-condition is zero during the first prediction or a detached previous $\hat{x}_0$ .         |
| Input conv   | 3x3 conv                                            | [B,10,256,256]                                      | [B,48,256,256]  | First image-stream feature map.                                                                 |
| Down level 0 | 2 x ResBlock(48)                                    | [B,48,256,256]                                      | [B,48,256,256]  | Each ResBlock uses FiLM and adds $p_{256}$ ; stores two 256 skips.                              |
| Downsample 0 | stride-2 conv                                       | [B,48,256,256]                                      | [B,48,128,128]  | Stores one pre-downsample skip.                                                                 |
| Down level 1 | ResBlock 48->96, ResBlock 96->96                    | [B,48,128,128]                                      | [B,96,128,128]  | Each ResBlock uses FiLM and adds $p_{128}$ ; stores two 128 skips.                              |
| Downsample 1 | stride-2 conv                                       | [B,96,128,128]                                      | [B,96,64,64]    | Stores one pre-downsample skip.                                                                 |
| Down level 2 | ResBlock 96->192, ResBlock 192->192                 | [B,96,64,64]                                        | [B,192,64,64]   | Each ResBlock uses FiLM and adds $p_{64}$ ; stores two 64 skips.                                |
| Downsample 2 | stride-2 conv                                       | [B,192,64,64]                                       | [B,192,32,32]   | Stores one pre-downsample skip.                                                                 |
| Down level 3 | 2 x ResBlock(192) + SelfAttn + CrossAttn after each | [B,192,32,32]                                       | [B,192,32,32]   | Each ResBlock uses FiLM and adds $p_{32}$ ; x-token cross-attention is active at 32 resolution. |
| Middle       | ResBlock + SelfAttn + CrossAttn + ResBlock          | [B,192,32,32]                                       | [B,192,32,32]   | Bottleneck; x-token cross-attention is always present here.                                     |
| Up level 3   | 2 x concat skip + ResBlock + SelfAttn + CrossAttn   | [B,192,32,32]                                       | [B,192,32,32]   | Consumes two 32 skips; each ResBlock adds $p_{32}$ .                                            |
| Upsample 3   | nearest upsample + 3x3 conv                         | [B,192,32,32]                                       | [B,192,64,64]   | Moves to 64.                                                                                    |
| Up level 2   | 3 x concat skip + ResBlock                          | [B,192,64,64]                                       | [B,192,64,64]   | Consumes downsample skip plus two 64 skips; each ResBlock adds $p_{64}$ .                       |
| Upsample 2   | nearest upsample + 3x3 conv                         | [B,192,64,64]                                       | [B,192,128,128] | Moves to 128.                                                                                   |
| Up level 1   | 3 x concat skip + ResBlock                          | [B,192,128,128]                                     | [B,96,128,128]  | Consumes downsample skip plus two 128 skips; each ResBlock adds $p_{128}$ .                     |
| Upsample 1   | nearest upsample + 3x3 conv                         | [B,96,128,128]                                      | [B,96,256,256]  | Moves to 256.                                                                                   |
| Up level 0   | 3 x concat skip + ResBlock                          | [B,96,256,256]                                      | [B,48,256,256]  | Consumes downsample skip plus two 256 skips; each ResBlock adds $p_{256}$ .                     |
| Output head  | GroupNorm + SiLU + 3x3 conv                         | [B,48,256,256]                                      | [B,1,256,256]   | Predicts $v$ by default.                                                                        |

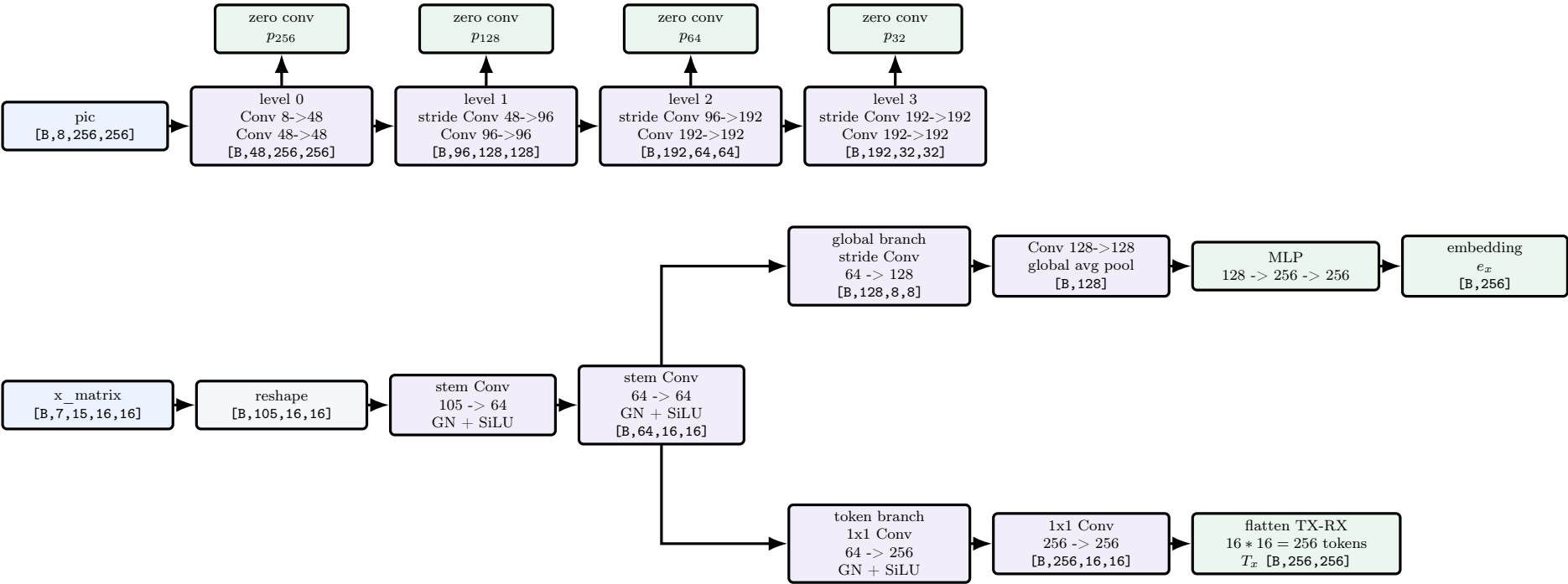


Core Blocks



| Block               | Stack rule in code                                                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ResBlock            | GroupNorm + SiLU + Conv, then FiLM-conditioned GroupNorm, SiLU + Dropout + Conv, plus identity or 1x1 skip. Every ResBlock receives $c = e_x + e_t$ .        |
| PicAdapter add      | After every ResBlock, the U-Net adds the PicEncoder zero-conv feature matching the current level. It is bilinearly resized only if the spatial size differs. |
| AttentionBlock      | Self-attention over flattened $H \times W$ locations. In the base48 config it is used at resolution 32 and in the middle block.                              |
| CrossAttentionBlock | Queries come from normalized U-Net spatial tokens. Keys and values come from TX-RX x tokens. It is used at resolution 32 and in the middle block.            |
| Downsample          | 3x3 stride-2 convolution. Channels stay the same; resolution halves.                                                                                         |
| Upsample            | nearest-neighbor upsample by 2, then 3x3 convolution. Channels stay the same; resolution doubles.                                                            |
| Skip concatenation  | Decoder ResBlocks concatenate the current feature with the latest saved encoder skip tensor along channel dimension before the ResBlock.                     |

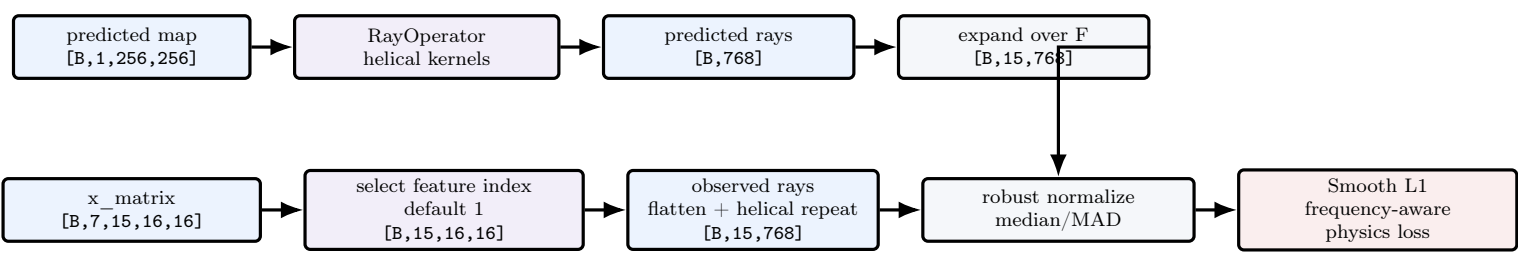
Condition Encoders



| Encoder output                     | Where it is used                                                                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $e_x$ [B, 256]                     | Added to the timestep embedding $e_t$ and passed to every ResBlock FiLM linear layer.                                                                                              |
| $T_x$ [B, 256, 256]                | Used as keys and values in CrossAttentionBlock at 32 resolution and in the middle block. This keeps TX-RX spatial structure instead of collapsing all x information to one vector. |
| $p_{256}, p_{128}, p_{64}, p_{32}$ | Added after each ResBlock at the matching U-Net scale. Zero-conv outputs start from zero influence and learn the coarse-map residual control during training.                      |

Diffusion, Self-Conditioning, and Physics Equations

| Step                 | Formula and tensor shape                                                                                                                                                                                                                              |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Forward noising      | $y_t = \sqrt{\bar{\alpha}_t}y_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$ , where $y_0, y_t, \epsilon$ are [B,1,256,256].                                                                                                                                  |
| Self-condition input | During training, with probability <b>self_condition_prob</b> =0.5, run one detached prediction and feed $\hat{x}_0^{prev}$ back into the final prediction. Otherwise use zeros. During sampling, each denoising step feeds the previous $\hat{x}_0$ . |
| Model image input    | <code>model_input = concat(<math>y_t, \hat{x}_0^{prev}, pic</math>)</code> gives [B,10,256,256].                                                                                                                                                      |
| Condition encoders   | <code>XMatrixEncoder(<math>x</math>) = (<math>e_x, T_x</math>)</code> with $e_x$ [B,256] and $T_x$ [B,256,256]. <code>PicEncoder(<math>pic</math>)</code> returns four adapter features.                                                              |
| Network prediction   | $\hat{v}$ = <code>ConditionalUNet(model_input, <math>t, e_x, T_x, pic</math>)</code> with output [B,1,256,256].                                                                                                                                       |
| Training target      | With <b>prediction_type=v_prediction</b> : $v = \sqrt{\bar{\alpha}_t}\epsilon - \sqrt{1 - \bar{\alpha}_t}y_0$ .                                                                                                                                       |
| Clean prediction     | $\hat{x}_0 = \sqrt{\bar{\alpha}_t}y_t - \sqrt{1 - \bar{\alpha}_t}\hat{v}$ , clamped to [0,1] for sampling and self-conditioning.                                                                                                                      |
| Main loss            | $L_{diff}$ = <code>MSE(<math>\hat{v}, v</math>)</code> plus configured auxiliary terms such as L1/SSIM, $\hat{x}_0$ L1, TV, range, and optional physics consistency.                                                                                  |
| Sampling             | DDPM/DDIM starts from $y_T \sim \mathcal{N}(0, I)$ with shape [B,1,256,256]. Each reverse step reuses the same pic and x_matrix conditions and updates self-condition with the current $\hat{x}_0$ .                                                  |



| Configuration note | Meaning                                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Main architecture  | This PDF uses <b>dataset_a_256_base48.yaml</b> , the current default 256-resolution diffusion architecture.                  |
| Base64 difference  | <b>dataset_a_256_base64.yaml</b> keeps the same topology but changes the channel sequence to 64,128,256,256.                 |
| Removed old paths  | The old global-embedding-only path, frequency-mean RayOperator path, and deleted debug configs are not part of this diagram. |